

Standard Operating Procedure (SOP):

Bi-Propellant Regeneratively Cooled Rocket Engine - Forward Unto Dawn

Team Name:	Harvard Undergraduate Aerospace Collective – Rocketry
Course/Student Group and Staff Project Coordinator name:	David Clarke
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Date:	9/1/2026

Location: (list all locations)	Type of space needed (maker space, wet lab, biolab, outside, drop testing, pool/water, etc.)	Space needs available in ALL's indoor teaching spaces (yes/no)	Have you confirmed with your staff project coordinator that you are approved to use the spaces listed (yes/no for each space type)	If using any space outside of ALL's indoor teaching, this must be reviewed by ALL staff team for approval and in some cases, a project proposal will need to be submitted to SEAS for review. If in SEAS buildings or SEAS outside spaces are need submit a request to SEAS Service Center: servicecenter@seas.harvard.edu
Project assembly	LL1 Undergraduate Lounge	No	Yes	
Project initial testing/proof of concept	Outside (Parking Lot or Grass Field)	No	No	
Project final testing	Off-Campus (Stow Island Air Park or Avon Fields)	No	No	
Additional testing stage:	N/A	No	No	

SECTION 1 – PROJECT SUMMARY

Please summarize your project, including its purpose and goals, then write out a step-by-step procedure of your testing proposal, experiment, proposed work, and fill out the safety assessment sections accordingly. If you plan to conduct iterative testing, please consider and explain how that may impact your safety plan.

- Forward Unto Dawn is a 250 pound-force regeneratively cooled liquid bi-propellant rocket engine, featuring nitrous oxide and ethanol as propellants. The purpose of this project is to successfully static-fire a liquid rocket engine, build institutional knowledge in rocket propulsion, and validate regenerative cooling as a design choice for future engines.

Asterisks indicate information that needs to be filled out in Section 2

Project initial assembly protocol

- List of supplies/items/environmental samples*
 - Project Initial Testing (Cold Flow): Liquid CO₂ (20lb), pressurized industrial-grade nitrogen (size K canister)
 - Project Final Testing (Static Fire): Liquid N₂O (20lb), pressurized industrial-grade nitrogen (size K canister), 200-proof anhydrous ethanol (8L), distilled water (4L)
- Sample collection plan and transit logistics
 - There is no sample collection
 - Project Initial Testing (Cold Flow): Supplies will be manually transported by team members
 - Project Final Testing (Static Fire): Supplies will be transported via vehicle to the static fire site
- Storage needs and identified storage location (chemicals, materials, hardware, ...)
 - 20 lb liquid N₂O, 20lb liquid CO₂ tank, size K pressurized nitrogen canister, 8L anhydrous ethanol
 - Compressed gas storage - Gas Cabinets (We currently don't have a storage location and are assessing options and recommendations)
 - Flammable Liquid Storage - LL1 Undergraduate Lounge / REEF Makerspace
 - All other materials - LL1 Undergraduate Lounge Club Room
- Plan for assembly
 - In-house manufacturing done in the Physics/SEAS Instructional Machine Shop
 - Engine assembly and integration done in the LL1 Undergraduate Lounge
 - Post engine manufacturing and assembly, we will connect the engine to our feed system and structures/test stand using bolts, NPT fittings, and other hardware.
 - Adapters will be needed for whichever kind of valve the supplied gas tanks have, and then the tanks can be connected to the rest of our feed system.
 - Most assembly will happen within the undergraduate lounge, but final assembly (connecting the tanks, engine) will be done onsite.
- PPE*
 - Manufacturing: Safety glasses, Closed-toe shoes, Nitrile Gloves
 - Cold Flow: Safety Glasses, Closed-toe shoes, water resistant gloves, Non-cotton clothing

- Static Fire: Safety Glasses, Closed-Toe Shoes, water resistant gloves, Non-cotton clothing, Nitrile Gloves
- Equipment to be used*
 - Manufacturing: Lathe, Mill, Drill Press, Hand Tools
- What are you trying to achieve with the initial testing protocol?
 - Cold Flow: The objective of this test will be to verify that there is sufficient atomization and impingement and to compare the spray angle to our predictions, as well as finding the manifold filling times for our mock oxidizer and mock fuel, giving us the data required to size the injector properly and to verify that there are no leaks present in the system.
- What are you trying to achieve with the final testing protocol?
 - Static Fire: To reach sonic and super-sonic flow, to verify cooling methods, to have successful ignition for a 2-second and 5-second burn time, to collect data (pressure and temperature) for comparison against simulations and expected results

Project initial testing

- Transportation and storage plan if off-site or outside of ALL spaces
 - Transportation: All necessary materials will be taken to the outside testing area (Parking lot or field) manually by team members.
 - Storage: All storage will be within SEAs
- Data collection plan (what data needs to be collected, and what equipment and spaces are going to be used)
 - We are looking to collect data such as the fuel insert pressure, fuel outlet pressure, chamber pressure, tank weights before and after, visual confirmation of flow.
 - To do this we are creating software for opening and closing our valves remotely, as well as abort logic to automatically shut off valves in case of overpressurization or excessive pressure drops.
 - First tests will be separate flows of the mock fuel (water), and the mock oxidizer (CO₂), in order to verify leak-free paths and injector orifice sizes, and to find the manifold filling times for our mock oxidizer and mock fuel.
 - Secondary tests will be combined flow of mock fuel and mock oxidizer, in order to verify impingement and atomization, as well as spray angle and mixing.
 - Throughout this cold flow testing procedure we will also verify our software, making sure that we notice any irregularities before we ever use flammable materials.
- Emergency planning (as needed: know the location of eye wash, safety shower, spill kit, other resources, ...)
 - Preventive: Staying a proper distance from the physical test for all team members
 - Emergency Protocol: Integrated abort procedures, Emergency Shut off Valve for gases

Project final testing (if different from initial testing)

- Transportation and storage plan if off-site or outside of ALL spaces
 - All necessary materials will be transported to the off-campus testing area via vehicular transport by team members.
- Data collection plan (what data needs to be collected, and what equipment and spaces are going to be used)
 - Static fire testing will be conducted in order to verify ignition, fuel insert pressure, fuel outlet pressure, chamber pressure, tank weights before and after, visual confirmation of flow, and visual confirmation of regenerative cooling efficacy.
 - We will run dress rehearsals of our valve software before testing with gasses, and we will test our abort procedures.

- Our first test will be an igniter test, to verify that we can ignite our a-class solid rocket motors for the simulated amount of time
- Our secondary test will be a 2-second hotfire, where we will start our igniter, open our oxidizer and fuel valves with the timing we measured during cold flow testing, and ensure that fuel and oxidizer reach the combustion chamber at the same time. The purpose of this test is to verify ignition, sonic flow, choked flow, and supersonic flow of the combustion gasses (essentially, verify that the rocket engine behaves as expected). The reason for such a short fire is to ensure that if the regenerative cooling does not behave as expected, we will not have burn-through of the engine walls.
- Our final tests will be 5-second hotfires, where we will do everything done in our 2-second hotfires, just for longer. The purpose of these tests is to verify that our engine can endure sustained fires without succumbing to the brutal laws of thermodynamics.
- Emergency planning
 - Preventive: Staying a proper distance from the physical test for all team members, watering the ground area of the static fire site
 - Emergency Protocol: Integrated abort procedures, Emergency Shut off Valve for gases, Fire extinguisher, Two 1-gallon water containers

SECTION 2 – HAZARD ASSESSMENT

Chemical Safety Resources:

- [Safety Data Sheets](#)
- [Lab Safety Guidelines](#)
- [Radioactive Materials](#)

Hazardous Chemicals Used	 Acutely Toxic	 Corrosive	 Flammable	 Oxidizer	 Irritant	 Explosive	 Health Hazard	 Compressed Gas	 Environmental Hazard
Liquid Nitrous Oxide https://jr.chemwatch.net/chemwatch.web/home	☐	☐	☐	☒	☒	☒	☐	☒	☐
Liquid Carbon Dioxide https://jr.chemwatch.net/chemwatch.web/home	☐	☐	☐	☐	☐	☐	☐	☒	☐
Anhydrous Ethanol https://jr.chemwatch.net/chemwatch.web/home	☐	☐	☒	☐	☒	☐	☐	☐	☐
Pressurized Nitrogen https://jr.chemwatch.net/chemwatch.web/home	☐	☐	☐	☐	☐	☐	☐	☒	☐
Lithium grease https://lentusllc.com/wp-content/uploads/2026/06/20574-RED-N-TACKY%C2%AF-Grease-SDS-CA-FN-2026.01.08.pdf	☐	☐	☐	☐	☒	☐	☐	☐	☐

Identify chemical storage location (lab storage guideline):

- Compressed gas storage: Gas Cabinets (We currently don't have a storage location and are assessing options and recommendations)
 - Liquid Nitrous Oxide
 - Liquid Carbon Dioxide
 - Pressurized Nitrogen Cannister

- Flammable Liquid Storage - LL1 Undergraduate Lounge / REEF Makerspace
 - Anhydrous Ethanol

Biological Safety Resources: N/A

- [BL1 Practices](#)
- [BL2 Practices](#)
- [BL2+ Practices](#)

Biological Materials Planned Use	Safety Level (BL1/BL2), source of material, and other handling considerations
N/A	N/A
N/A	N/A

Physical Hazards Resources: • Hand and Portable Power Tools • Electrical Safety • Machine Shop Safety • Makerspace Safety • Fall Protection • Hearing Conservation	Radiation Protection Resources: • Radioactive Materials • Lasers • XRays • High-Powered Magnets • Unenclosed UV sources
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Link to Supply List/Bill of Materials:

<https://docs.google.com/spreadsheets/d/1Zr2hxnZ44px2q-YFJBDsu4nz31eBQ9WbfY7kP7nnMEs/edit?pli=1&gid=2082117765#gid=2082117765>

Supply list (wood, PVC, metal bike frame, Li battery, pressurized CO2)	
Liquid CO2 (20lb) and the tank that goes with it	
Liquid N2O (20lb) and the tank that goes with it	
Pressurized industrial-grade nitrogen (size K canister)	
200-proof anhydrous ethanol (8L)	
Distilled water (4L)	

Equipment to be Used (ex. lathe, hand tools, welding, laser cutter, 3-D printer)	Site specific training status – specify training required, and if it has been completed, scheduled, or not yet determined	Risk profile
Lathe	Manual Lathe Training	Medium
Mill	Manual Mill Training	Medium
Drill Press	Shop Orientation and Basic Training	Low
Hand Tools	Machine Shop and Makerspace Safety	Low

Waste	
Generated	☐ Yes ☒ No
Disposal of waste	☐ Hazardous liquid ☐ Hazardous solid ☐ biological liquid ☐ Biological solid ☐ Sink ☐ Trash ☐ Disposal Restricted ☐ Other Regulated Recyclables (Li batteries, textiles, etc., describe: _____)
Do you have an approved waste disposal plan with your project coordinator?	☐ Yes ☐ No – must establish before materials are ordered and project assembly begins

PPE (Personal Protective Equipment) guides

- Safety Data Sheets
- General Lab Safety PPE
 - By [Type of PPE](#)
 - By [Task or Activity \(chemical, biological, radiation\)](#)
 - Glove guide

Please specify the PPE needed for each phase of your project by choosing from the list below

Location: (list all locations)	Gloves	Eye Protection	Body and Respiratory Protection
Project Assembly	☒ Nitrile ☐ Latex ☐ Neoprene ☐ Insulated ☐ Other: _____	☐ Goggles ☐ Face shield ☒ Safety Glasses ☐ Other: _____	☐ Lab coat ☐ Apron ☐ Flammable Lab coat ☐ N95 respirator or dust mask ☒ None ☐ Other: _____
Project initial testing/proof of concept	☐ Nitrile ☐ Latex ☐ Neoprene ☒ Insulated ☐ Other: _____	☐ Goggles ☐ Face shield ☒ Safety Glasses ☐ Other: _____	☐ Lab coat ☐ Apron ☐ Flammable Lab coat ☐ N95 respirator or dust mask ☒ None ☐ Other: _____
Project final testing	☒ Nitrile ☐ Latex ☐ Neoprene ☒ Insulated ☐ Other: _____	☐ Goggles ☐ Face shield ☒ Safety Glasses ☐ Other: _____	☐ Lab coat ☐ Apron ☐ Flammable Lab coat ☐ N95 respirator or dust mask ☒ None ☐ Other: _____
Additional testing stage(s):	☐ Nitrile ☐ Latex ☐ Neoprene ☐ Insulated ☐ Other: _____	☐ Goggles ☐ Face shield ☐ Safety Glasses ☐ Other: _____	☐ Lab coat ☐ Apron ☐ Flammable Lab coat ☐ N95 respirator or dust mask ☐ None ☐ Other: _____

Administrative and Engineering Controls

Please specify the safety controls needed for each phase of your project by choosing from the list below

	Air quality	Work Area	System Controls and Signage	Clean/Decontaminate Work Area
Project Assembly Anticipated location:	☐ Fume Hood ☐ Biosafety Cabinet ☐ Snorkel ☐ Filter (isoflurane)	☐ Fume Hood ☐ Biosafety Cabinet ☐ Bench ☒ Core/facility/shop _____ ☐ Other (e.g. address of offsite location): _____	☐ Pressure relief valve ☒ Emergency stop button ☐ Other _____	☐ Water & soap ☐ 70% Ethanol ☐ __% Bleach ☒ None ☐ Specify _____

Project initial testing/proof of concept Anticipated Location:	☐ Fume Hood ☐ Biosafety Cabinet ☐ Snorkel ☐ Filter (isoflurane)	☐ Fume Hood ☐ Biosafety Cabinet ☐ Bench ☐ Core/facility/shop _____ ☒ Other (e.g. address of offsite location): SEC Field/Parking Lot	☒ Pressure relief valve ☒ Emergency stop button ☐ Other _____	Water & soap ☐ 70% Ethanol ☐ __% Bleach ☐ None ☐ Specify _____
Project final testing Anticipated location:	☐ Fume Hood ☐ Biosafety Cabinet ☐ Snorkel ☐ Filter (isoflurane)	☐ Fume Hood ☐ Biosafety Cabinet ☐ Bench ☐ Core/facility/shop _____ ☒ Other (e.g. address of offsite location): Avon Farm Fields, CT	☒ Pressure relief valve ☒ Emergency stop button ☐ Other _____	Water & soap ☐ 70% Ethanol ☐ __% Bleach ☐ None ☐ Specify _____
Additional testing stage(s) Anticipated Location:	☐ Fume Hood ☐ Biosafety Cabinet ☐ Snorkel ☐ Filter (isoflurane)	☐ Fume Hood ☐ Biosafety Cabinet ☐ Bench ☐ Core/facility/shop _____ ☐ Other (e.g. address of offsite location): _____	☐ Pressure relief valve ☐ Emergency stop button ☐ Other _____	Water & soap ☐ 70% Ethanol ☐ __% Bleach ☐ None ☐ Specify _____

Required [Training](#) based on hazards identified above:

Training	Yes
General Laboratory Safety	No
Laboratory Biosafety	No
Laser Safety	No
Radioactive Materials Safety	No
Machine Shop and Makerspace Safety	Yes
Respiratory Protection	No
Other: Hot Fire Safety	Yes

- [Laboratory Chemical Storage Guide](#)
- [Hazardous Waste Procedures](#)
- [EHS Lab Emergency Response Guide](#);